

2022 Commodity Flow Survey
Public Use Microdata Sample File – Data Users Guide
Issued: January 2026

Introduction

This Data Users Guide describes the Public Use Microdata Sample (PUMS) File created from the data collected for the 2022 Commodity Flow Survey (CFS). It provides a brief description of the CFS, outlines the differences between the CFS data used to create the published estimates and the records available in the CFS PUMS, and explains how to create estimates and measures of sampling variability using the CFS PUMS data. Readers interested in understanding or analyzing the CFS PUMS should also familiarize themselves with the content of the **2022 CFS Methodology**¹ to obtain an understanding of the survey, questionnaire, sample design, and other topics relevant to the CFS.

Comments from users regarding the content and usefulness of this data product are appreciated. Contact the CFS staff by email (ERD.CFS@census.gov) or phone (301-763-2108) with your feedback.

About the Commodity Flow Survey

The CFS is a joint effort of the Bureau of Transportation Statistics (BTS), U.S. Department of Transportation, and the U.S. Census Bureau, U.S. Department of Commerce. The survey is the primary source of national and subnational (state and metropolitan area) data on domestic freight shipments by establishments located in the 50 U.S. states and the District of Columbia and classified in Mining, Manufacturing, Wholesale, and selected Retail and Service trade industries, as defined by the 2017 North American Industry Classification System (NAICS)². Auxiliary establishments, such as warehouses and central administrative offices with shipping activity that support the operations of large multi-establishment companies, are also included.

The survey produces estimates on the type, origin and destination, value, weight, mode of transportation, and distance of commodities shipped. Conducted every five years as part of the Economic Census³, the CFS provides a modal picture of national freight flows and represents the only publicly available source of commodity flow data for the highway mode. The CFS has been conducted every five years since 1993, with the most recent cycle in 2022.

PUMS-based tabulations may differ substantially from published CFS estimates and therefore should not be treated as substitutes for the official CFS estimates. For custom data needs, the Census Bureau offers special tabulations prepared by CFS staff using the full internal data. To discuss exact specifications for the type and format of the request, contact the CFS staff by email (ERD.CFS@census.gov) or phone (301-763-2108).

¹ <<https://www.census.gov/programs-surveys/cfs/technical-documentation/methodology.html>>

² Establishments classified in transportation (other than freight trucking and warehousing), construction, and most retail and services industries are excluded. Together, these sectors have several million establishments, few of which are likely to have significant shipping activity. For more information on out-of-scope industries, refer to “Industry Coverage” in the **2022 CFS Methodology**, linked above in footnote 1.

³ <<https://www.census.gov/programs-surveys/economic-census.html>>

The CFS PUMS Contents

The CFS PUMS is made available as a comma-separated values (CSV) file and includes 18 variables for a sample of usable⁴ shipment records collected by the CFS. The file contains a total of 37,576,546 shipments from approximately 50,000 responding establishments. Each shipment record includes the following information (see APPENDIX A **Table A1** for variable names, detailed definitions, and coding):

- Shipment identifier
- Shipment origin
 - State
 - Metropolitan area
- Shipment destination (U.S.)
 - State
 - Metropolitan area
- Industry sector of the shipper
- Type of commodity
- Mode of transportation
- Shipment value (in dollars)
- Shipment weight (in pounds)
- Great-circle distance between shipment origin and U.S. destination (in miles)
- Temperature-controlled indicator
- Export shipment indicator
- Final export destination code (if applicable)
- Hazardous material code
- Shipment tabulation weighting factor – used to expand CFS PUMS shipment-level numeric data (i.e., shipment value, shipment weight, shipment distance) to represent the total population of in-scope U.S. shipments in 2022. This factor is the product of seven component survey weights, disclosure avoidance noise factors, and microdata sampling adjustment factors. Details on the seven component survey weights are provided in the **2022 CFS Methodology**, linked in footnote 1.

Note: The shipment tabulation weighting factor (WGT_FACTOR) assigned to a shipment is also an estimate of the number of shipments of the type represented by that PUMS record⁵.

*Note: What constitutes a shipment is determined by the respondent; there are no minimum or maximum limits on shipment weight, value, volume, or size. The only requirements are that a shipment has a single destination and requires one trip. A single shipment may consist of multiple pieces or be transported using multiple vehicles, such as unit trains or truck convoys, but must have only one destination. For additional details on the guidance provided to CFS respondents, refer to the **2022 CFS Online Survey Instructions** at <www.census.gov/programs-surveys/cfs/technical-documentation/questionnaires.html>.*

⁴ A *usable* shipment is one that contributes to tabulations. A shipment may be considered *unusable* if the respondent fails to provide the value, weight, commodity, or destination for a sampled shipment and the missing information could not be imputed or otherwise obtained. For more information on usable shipments, refer to “Estimation” in the **2022 CFS Methodology**, linked above in footnote 1.

⁵ Note: A small number of shipments have weighting factors < 1. This may occur when the tabulated revenue data from the 2022 Economic Census is less than the CFS weighted shipment value (computed using all prior weighting factors). Refer to “Estimation” in the **2022 CFS Methodology**, linked in footnote 1, for a description of the weighting procedure.

Differences Between Published CFS Estimates and CFS PUMS Tabulations

To protect the confidentiality of the CFS respondents, the CFS uses noise-infusion when producing estimates of shipment value and shipment weight. Details relating to this disclosure avoidance technique may be found in the “Disclosure Avoidance” section of the **2022 CFS Methodology**, linked in footnote 1, above.

For the 2022 CFS PUMS, additional measures were taken to protect the confidentiality of the data of the CFS respondents. These additional measures to protect confidentiality include:

- **Sampling.** Shipments were sampled within each quarter within an establishment (establishment-quarter) when the number of shipments exceeded a specified threshold. The sample design is a stratified simple random sample without replacement. The strata are defined by the cross-classifications of establishment, quarter, destination CFS area, 2-digit SCTG code, mode of transportation, export country, temperature control (yes/no), and HAZMAT code. Sample sizes were allocated to strata using proportional allocation based on the cross-classifications of destination CFS area, 2-digit SCTG code, mode of transportation, export country, temperature control (yes/no), and HAZMAT code within each establishment-quarter. A poststratification adjustment was then applied so that estimates of total shipment value based on the sampled shipments matched the totals that would have been obtained without sampling. Poststratification cells were defined using the same cross-classifications as the sampling strata.

Note: The effects of the shipment sampling and poststratification are reflected in the PUMS tabulation weighting factor (WGT_FACTOR). When applied, WGT_FACTOR expands the sampled shipments to represent the full population of usable shipments used to produce the 2022 CFS estimates.

- **Truncation and topcoding.** Shipments with value exceeding 30,000,000 were truncated to equal 30,000,000. In addition, extremely large shipment values and shipment weights were subsequently top-coded. Extremely large shipment weighting factors were top-coded to 3,588,000.

Note: Following truncation, shipment weight (SHIPMT_WGHT) was adjusted to preserve each shipment’s value-to-weight ratio. This process may result in underestimation of shipment value and/or shipment weight. Therefore, when a direct estimate is available in one of the published 2022 CFS tables⁶, users are encouraged to use the published estimate rather than one derived from the PUMS file.

Note: Following topcoding, adjustments were made so that, for each shipment, the products of shipment value (SHIPMT_VALUE) and shipment weight (SHIPMT_WGHT) with the tabulation weighting factor (WGT_FACTOR) remained approximately equal to their values prior to these changes.

- **Application of noise.** Additional noise was applied to the shipment value and shipment weight.

Note: Following the application of additional noise, adjustments were made so that, for each shipment, the products of shipment value (SHIPMT_VALUE) and shipment weight (SHIPMT_WGHT) with the tabulation weighting factor (WGT_FACTOR) remained approximately equal to their values prior to these changes.

- **Rounding.** Shipment value, shipment weight, and great-circle distance quantities were rounded to the nearest integer. The tabulation weighting factors were rounded to the nearest tenth.
- **Confidential collapsing.** For approximately 100,000 shipments, the detail for origin and/or commodity was reduced. For example, a shipment origin may have been changed from “Chicago, IL CFS Area” to “Illinois”.

Note: This collapsing may result in underestimation at detailed origin geography or commodity levels as some shipments were recoded into higher-level aggregates. Therefore, when a direct estimate is

⁶ <<https://www.census.gov/data/tables/2022/econ/cfs/aff-2022.html>>

available in one of the published 2022 CFS tables⁶, users are encouraged to use the published estimate rather than one derived from the PUMS file.

- **Additional data coarsening.** For certain categorical variables, published categories were combined to reduce the risk of identifying respondents based on rare characteristics. The following CFS PUMS variables were recoded due to disclosure concerns and differ from the categories in the published CFS tables.
 1. Industry sector of shipper (SECTOR)
Published CFS tables include estimates tabulated by a shipper's 3- or 4-digit NAICS industry classification code. In the CFS PUMS, the shipper's NAICS industry is collapsed into seven 2-digit sector codes.
 2. Commodity code (SCTG)
Published CFS tables include estimates tabulated by a shipment's 2-, 3-, or 4-digit SCTG code. In the CFS PUMS, the SCTG is collapsed into 42 2-digit codes (or further reduced as described in **Confidential collapsing**).
 3. Hazardous material code (HAZMAT)
Published CFS tables include estimates tabulated by a shipment's UN/NA code⁷. In the CFS PUMS, the UN/NA code is collapsed into three HAZMAT indicator codes.

While implementing these measures, certain desirable properties of the original data were preserved to the extent possible:

- The value-to-weight ratio of individual shipments was maintained.
- When detail had to be reduced (as in **Confidential collapsing**), commodity detail was reduced before geographic detail. All geography and commodity levels can be found in APPENDIX A **Table A2** and **Table A3**, respectively.
- Estimates produced from the CFS PUMS would remain close to the published estimates.

Estimates produced using the 2022 CFS PUMS have the same issues with comparability to prior surveys as the published 2022 estimates. Data users should be cautious when comparing any 2022 estimates with CFS estimates from prior years. See the "Comparability of Estimates" section of the **2022 CFS Methodology**, linked in footnote 1, for a discussion of these comparability issues.

The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data (Project No. P-7504831, Disclosure Review Board (DRB) approval number: CBDRB-FY26-014).

Differences between the 2017 CFS PUF and the 2022 CFS PUMS

The 2022 CFS PUMS differs from the 2017 CFS PUF in several important ways:

1. Respondents in 2022 could, for the first time, provide all shipments for a given week rather than a sample (refer to "Third Stage – Shipment Selection" in the **2022 CFS Methodology**, linked in footnote 1). Because this option significantly increased the number of reported shipments, the 2022 PUMS includes only a sample of the total usable shipments used in producing the 2022 CFS estimates. In contrast, the 2017 file included all usable shipments and was referred to as a Public Use File (PUF).
2. In 2017, the mode variable (MODE) was used in confidential collapsing and therefore the 2017 file included additional, aggregated mode codes. In 2022, mode was not used in confidential collapsing.
3. The variable QUARTER was included in 2017 but is no longer provided on the 2022 CFS PUMS.

⁷ UN/NA is the United Nations or North American code for hazardous materials.

4. The 2017 CFS published estimates of routed mileage for shipments. The 2022 CFS did not calculate routed mileage and therefore this variable no longer appears in the CFS PUMS (refer to “Mileage Calculation Changes” in the **2022 CFS Methodology**, linked in footnote 1). Great-circle distance (SHIPMT_DIST_GC) continues to be provided, as it was in 2017.
5. The 2017 file provided the publication NAICS code for each shipment. In 2022, shipments are categorized by NAICS sector, which is less detailed than the publication NAICS codes.

Creating Estimates with the CFS PUMS

The CFS design includes three stages of sampling, adjustments for survey non-response, and post-stratification to Economic Census totals; therefore, *tabulation weighting factors **must** be incorporated when producing estimates with the CFS PUMS data.* Additionally, an estimate of sampling variance should be derived for each associated weighted estimate.

How to estimate total value and total tonnage with the CFS PUMS

To create estimates of the total shipment value (in dollars), multiply each shipment’s value by its tabulation weighting factor and then sum the results over all shipments in the domain of interest:

$$Total\ Shipment\ Value_D = \sum_{i \in D} WGT_FACTOR_i \cdot SHIPMT_VALUE_i, \quad [1]$$

where D is the domain of interest, WGT_FACTOR is the shipment tabulation weighting factor, $SHIPMT_VALUE$ is the value of the shipment, and i indexes the shipments in domain D . For example, to estimate the total value of shipments originating in Maryland, compute the quantity in [1] using shipments where $ORIG_STATE = 24$. Note that the domain does not need to be geographical. To instead estimate the total value of shipments of basic chemicals, compute the quantity in [1] using shipments where $SCTG = 20$.

Note: Published estimates of total shipment value are provided in millions of dollars. When comparing tabulations derived from the PUMS to published estimates, differences due to rounding should be considered.

Estimates of the total shipment tonnage can be calculated similarly by multiplying each shipment’s weight (divided by 2,000) by its tabulation weighting factor and then summing the results over all shipments in the domain of interest:

$$Total\ Shipment\ Tonnage_D = \sum_{i \in D} WGT_FACTOR_i \cdot [SHIPMT_WGHT_i / 2000], \quad [2]$$

where $SHIPMT_WGHT$ is the weight of the shipment (in pounds) and D , WGT_FACTOR , and i are as defined above. The division by 2,000 converts pounds to tons.

Note: Published estimates of total shipment weight are provided in thousands of tons. When comparing tabulations derived from the PUMS to published estimates, differences due to rounding should be considered.

How to estimate average distance (GCD) per shipment with the CFS PUMS

Estimates of the average distance (GCD) per shipment for a domain of interest can be calculated by taking the weighted sum of shipment distances and dividing by the sum of the tabulation weighting factors:

$$\text{Average Distance per Shipment}_D = \frac{\sum_{i \in D} (WGT_FACTOR_i \cdot SHIPMT_DIST_GC_i)}{\sum_{i \in D} WGT_FACTOR_i}, \quad [3]$$

where $SHIPMT_DIST_GC$ is the great-circle distance between the shipment origin and destination, and D , WGT_FACTOR , and i are as defined above. This produces the average distance (GCD) per shipment, weighted by the number of shipments represented by each CFS PUMS record.

How to estimate coefficients of variation (CVs) with the CFS PUMS

A generalized variance function (GVF) method was developed to allow users to compute coefficients of variation (CVs) for estimates derived from the CFS PUMS that include noise introduced by the disclosure avoidance procedures. This approach is necessary because, due to confidentiality concerns, the random group information required to compute CVs using the method of random groups cannot be included on the PUMS file⁸.

The basic idea of the GVF method is to specify a function (the GVF) that expresses the CV in terms of quantities that can be computed with the CFS PUMS and a set of unknown parameters. This function is incorporated into a linear regression model in which CVs computed using the method of random groups serve as the outcome variable, the PUMS-computable quantities are the covariates, and the unknown parameters correspond to the regression coefficients.

The regression coefficients are estimated by the Census Bureau and are provided to users. Using the quantities computed from the CFS PUMS together with the estimates of the regression coefficients, users can estimate CVs for their own tabulations.

Computing CVs for total value and total tonnage

For estimates of total value and total tonnage, the GVF is defined as:

$$\ln(CV_{total}) = a + b \ln(n) + c[\ln(n)]^2, \quad [4]$$

where $\ln$ denotes the natural logarithm (base e), CV_{total} is the CV (expressed as a percent) of the total estimate, n is the number of shipments (i.e., PUMS records) used to compute the estimate, and a , b , and c are the regression coefficients provided in **Table 1**. below.

Accordingly, the CV of an estimate of a total can be computed as:

$$CV_{total} = e^{a+b \ln(n)+c[\ln(n)]^2}. \quad [5]$$

*Note: Select the appropriate values of a , b , and c from **Table 1**. based on the type of total being estimated.*

⁸ The method of random groups is used to compute the CVs of the CFS estimates that are released to the public.

Computing CVs for average distance (GCD) per shipment

For estimates of average distance per shipment (ADPS), the GVF is defined as:

$$\ln(CV_{ADPS}) = a + b \ln(n) + c \ln(ADPS), \quad [6]$$

where $\ln$ denotes the natural logarithm (base e), CV_{ADPS} is the CV (expressed as a percent) of the estimate of average distance per shipment, n is the number of shipments (i.e., PUMS records) used to compute the estimate, $ADPS$ is the estimate of average distance per shipment, and a , b , and c are the regression coefficients provided in **Table 1.** below.

Table 1.

Regression Coefficients of the GVFs

Type of Estimate	a	b	c
Total value	3.905	0.011	-0.010
Total tonnage	3.837	0.048	-0.010
Average distance (GCD) per shipment	5.381	-0.056	-0.416

Accordingly, the CV of an estimate of average distance per shipment can be computed as:

$$CV_{ADPS} = e^{a+b \ln(n)+c \ln(ADPS)}. \quad [7]$$

*Note: Select the appropriate values of a , b , and c from **Table 1.** based on the type of total being estimated.*

Example of calculations using the CFS PUMS

In this section, calculations of the weighted PUMS estimate and associated CV are shown for the *total value of shipments originating in Maryland* (here, the domain D is defined as shipments where $ORIG_STATE = 24$). Note that the number of shipments on the CFS PUMS originating in Maryland is 612,317.

First, use [1] to calculate the weighted *total value* estimate by summing the product of each shipment's tabulation weighting factor with its value:

$$\hat{Y}_D = \sum_{i \in (ORIG_STATE=24)} WGT_FACTOR_i \cdot SHIPMT_VALUE_i = 198,598,788,295.$$

Next, use [5] to compute the associated CV for the *total value of shipments originating in Maryland*, selecting the values of a , b , and c from the first row of **Table 1.**

$$\begin{aligned} cv(\hat{Y}_D) &= e^{a+b \ln(n)+c[\ln(n)]^2} \\ &= e^{3.905+0.011 \ln(612,317)-0.010[\ln(612,317)]^2} \\ &= 9.74\% \end{aligned}$$

APPENDIX A: 2022 CFS Public Use Microdata Sample (PUMS) File Data Dictionary

Table A1.

Data Dictionary – File Layout

Field	Description	Valid Values	Type	Column Number
SHIPMT_ID	Shipment identifier <i>(Note: This variable is an integer assigned by randomly sorting shipment records and numbering them sequentially starting at 00000001).</i>	00000001 – 37576546	CHAR	1
ORIG_STATE	FIPS state code of shipment origin	<i>See Note (1)</i>	CHAR	2
ORIG_MA	Metro area of shipment origin	<i>See Note (1)</i>	CHAR	3
ORIG_CFS_AREA	CFS area of shipment origin <i>(Note: This variable is formed by concatenating ORIG_STATE and ORIG_MA, separated by a hyphen (e.g., 24-12580))</i>	<i>See Note (1)</i>	CHAR	4
DEST_STATE	FIPS state code of shipment destination	<i>See Note (1)</i>	CHAR	5
DEST_MA	Metro area of shipment destination	<i>See Note (1)</i>	CHAR	6
DEST_CFS_AREA	CFS area of shipment destination <i>(Note: This variable is formed by concatenating DEST_STATE and DEST_MA, separated by a hyphen (e.g., 01-142))</i>	<i>See Note (1)</i>	CHAR	7
SECTOR	Industry sector of shipper	21 = Mining, quarrying, and oil and gas extraction 31-33 = Manufacturing 42 = Wholesale trade 44-45 = Retail trade 48-49 = Transportation and warehousing 51 = Information 55 = Management of companies and enterprises	CHAR	8
SCTG	2-digit Standard Classification of Transported Goods (SCTG) code identifying the primary commodity transported in the shipment	<i>See Note (2)</i>	CHAR	9
MODE	Mode of transportation of the shipment	111 = For-hire truck 112 = Company-owned truck 113 = Customer pick-up 12 = Rail 131 = Inland water 132 = Great Lakes 133 = Deep sea 14 = Air (includes truck and air) 15 = Pipeline 16 = Other single modes 21 = Parcel, U.S.P.S. or courier 22 = Truck and rail	CHAR	10

Field	Description	Valid Values	Type	Column Number
		23 = Truck and water 24 = Rail and water 25 = Other multiple modes 30 = Unknown mode		
SHIPMT_VALUE	Value of the shipment (in dollars)	0 – 99,999,999	NUM	11
SHIPMT_WGHT	Weight of the shipment (in pounds)	0 – 999,999,999	NUM	12
SHIPMT_DIST_GC	Great-circle distance between the shipment origin and destination (in miles)	0 – 9,999	NUM	13
TEMP_CNTL_YN	Temperature controlled shipment indicator (yes/no)	Y = Yes N = No	CHAR	14
EXPORT_YN	Export shipment indicator (yes/no)	Y = Yes N = No	CHAR	15
EXPORT_CNTRY	Country of destination (for export shipments)	A = Asia and Oceania C = Canada E = Europe and Africa M = Mexico N = Not an export S = Rest of the Americas	CHAR	16
HAZMAT	Hazardous material (HAZMAT) code	H = Other HAZMAT N = Not HAZMAT P = Class 3 HAZMAT (flammable liquids)	CHAR	17
WGT_FACTOR	Shipment tabulation weighting factor <i>(Note: This factor is also an estimate of the total number of shipments represented by the PUMS file shipment.)</i>	0.0 – 9,999,999.9	NUM	18

Notes:

- (1) See **Table A2.** for the values and descriptions of valid metro areas, CFS areas, and states. For more information on the CFS areas, refer to “Geographic Strata” in the **2022 CFS Methodology**, linked above in footnote 1.
- (2) See **Table A3.** for the values and descriptions of valid SCTG commodity codes.

Table A2.

Data Dictionary – Origin and Destination Geographies

Metro area of shipment	State of shipment	CFS Area of shipment	MA Type	Description
ORIG_MA DEST_MA	ORIG_STATE DEST_STATE	ORIG_CFS_AREA DEST_CFS_AREA		
104	36	36-104	C	Albany-Schenectady, NY CFS Area
122	13	13-122	C	Atlanta-Athens-Clarke County-Sandy Springs, GA-AL CFS Area (GA Part)
142	01	01-142	C	Birmingham-Hoover-Talladega, AL CFS Area
148	25	25-148	C	Boston-Worcester-Providence, MA-RI-NH-CT CFS Area (MA Part)
	33	33-148	C	Boston-Worcester-Providence, MA-RI-NH-CT CFS Area (NH Part)
	44	44-148	C	Boston-Worcester-Providence, MA-RI-NH-CT CFS Area (RI Part)
160	36	36-160	C	Buffalo-Cheektowaga-Olean, NY CFS Area
172	37	37-172	C	Charlotte-Concord, NC-SC CFS Area (NC Part)
176	17	17-176	C	Chicago-Naperville, IL-IN-WI CFS Area (IL Part)
	18	18-176	C	Chicago-Naperville, IL-IN-WI CFS Area (IN Part)
178	21	21-178	C	Cincinnati-Wilmington-Maysville, OH-KY-IN CFS Area (KY Part)
	39	39-178	C	Cincinnati-Wilmington-Maysville, OH-KY-IN CFS Area (OH Part)
184	39	39-184	C	Cleveland-Akron-Canton, OH CFS Area
198	39	39-198	C	Columbus-Marion-Zanesville, OH CFS Area
204	48	48-204	C	Corpus Christi-Kingsville-Alice, TX CFS Area
206	48	48-206	C	Dallas-Fort Worth, TX-OK CFS Area (TX Part)
212	39	39-212	C	Dayton-Springfield-Kettering, OH CFS Area
216	08	08-216	C	Denver-Aurora, CO CFS Area
218	19	19-218	C	Des Moines-Ames-West Des Moines, IA CFS Area
220	26	26-220	C	Detroit-Warren-Ann Arbor, MI CFS Area
238	48	48-238	C	El Paso-Las Cruces, TX-NM CFS Area (TX Part)
258	18	18-258	C	Fort Wayne-Huntington-Auburn, IN CFS Area
260	06	06-260	C	Fresno-Madera-Hanford, CA CFS Area
266	26	26-266	C	Grand Rapids-Kentwood-Muskegon, MI CFS Area
268	37	37-268	C	Greensboro--Winston-Salem--High Point, NC CFS Area
273	45	45-273	C	Greenville-Spartanburg-Anderson, SC CFS Area
288	48	48-288	C	Houston-The Woodlands, TX CFS Area
294	18	18-294	C	Indianapolis-Carmel-Muncie, IN CFS Area
300	12	12-300	C	Jacksonville-St. Marys-Palatka, FL-GA CFS Area (FL Part)
312	20	20-312	C	Kansas City-Overland Park-Kansas City, MO-KS CFS Area (KS Part)
	29	29-312	C	Kansas City-Overland Park-Kansas City, MO-KS CFS Area (MO Part)
315	47	47-315	C	Knoxville-Morristown-Sevierville, TN CFS Area
324	22	22-324	C	Lake Charles-Jennings, LA CFS Area
332	32	32-332	C	Las Vegas-Henderson, NV CFS Area
348	06	06-348	C	Los Angeles-Long Beach, CA CFS Area
350	21	21-350	C	Louisville/Jefferson County-Elizabethtown-Bardstown, KY-IN CFS Area (KY Part)
368	47	47-368	C	Memphis-Forrest City, TN-MS-AR CFS Area (TN Part)
370	12	12-370	C	Miami-Port St. Lucie-Fort Lauderdale, FL CFS Area
376	55	55-376	C	Milwaukee-Racine-Waukesha, WI CFS Area
378	27	27-378	C	Minneapolis-St. Paul, MN-WI CFS Area (MN Part)
380	01	01-380	C	Mobile-Daphne-Fairhope, AL CFS Area
400	47	47-400	C	Nashville-Davidson--Murfreesboro, TN CFS Area
406	22	22-406	C	New Orleans-Metairie-Hammond, LA-MS CFS Area (LA Part)
408	09	09-408	C	New York-Newark, NY-NJ-CT-PA CFS Area (CT Part)
	34	34-408	C	New York-Newark, NY-NJ-CT-PA CFS Area (NJ Part)
	36	36-408	C	New York-Newark, NY-NJ-CT-PA CFS Area (NY Part)
416	40	40-416	C	Oklahoma City-Shawnee, OK CFS Area
420	31	31-420	C	Omaha-Council Bluffs-Fremont, NE-IA CFS Area (NE Part)
422	12	12-422	C	Orlando-Lakeland-Deltona, FL CFS Area

Metro area of shipment	State of shipment	CFS Area of shipment	MA Type	Description
ORIG_MA DEST_MA	ORIG_STATE DEST_STATE	ORIG_CFS_AREA DEST_CFS_AREA		
428	10	10-428	C	Philadelphia-Reading-Camden, PA-NJ-DE-MD CFS Area (DE Part)
	34	34-428	C	Philadelphia-Reading-Camden, PA-NJ-DE-MD CFS Area (NJ Part)
	42	42-428	C	Philadelphia-Reading-Camden, PA-NJ-DE-MD CFS Area (PA Part)
429	04	04-429	C	Phoenix-Mesa, AZ CFS Area
430	42	42-430	C	Pittsburgh-New Castle-Weirton, PA-OH-WV CFS Area (PA Part)
440	41	41-440	C	Portland-Vancouver-Salem, OR-WA CFS Area (OR Part)
	53	53-440	C	Portland-Vancouver-Salem, OR-WA CFS Area (WA Part)
450	37	37-450	C	Raleigh-Durham-Cary, NC CFS Area
464	36	36-464	C	Rochester-Batavia-Seneca Falls, NY CFS Area
472	06	06-472	C	Sacramento-Roseville, CA CFS Area
476	17	17-476	C	St. Louis-St. Charles-Farmington, MO-IL CFS Area (IL Part)
	29	29-476	C	St. Louis-St. Charles-Farmington, MO-IL CFS Area (MO Part)
482	49	49-482	C	Salt Lake City-Provo-Orem, UT CFS Area
484	48	48-484	C	San Antonio-New Braunfels-Pearsall, TX CFS Area
488	06	06-488	C	San Jose-San Francisco-Oakland, CA CFS Area
496	13	13-496	C	Savannah-Hinesville-Statesboro, GA CFS Area
500	53	53-500	C	Seattle-Tacoma, WA CFS Area
534	39	39-534	C	Toledo-Findlay-Tiffin, OH CFS Area
536	04	04-536	C	Tucson-Nogales, AZ CFS Area
538	40	40-538	C	Tulsa-Muskogee-Bartlesville, OK CFS Area
545	51	51-545	C	Virginia Beach-Norfolk, VA-NC CFS Area (VA Part)
556	20	20-556	C	Wichita-Winfield, KS CFS Area
10900	42	42-10900	M	Allentown-Bethlehem-Easton, PA-NJ CFS Area (PA Part)
12420	48	48-12420	M	Austin-Round Rock-Georgetown, TX CFS Area
12580	24	24-12580	M	Baltimore-Columbia-Towson, MD CFS Area
12940	22	22-12940	M	Baton Rouge, LA CFS Area
13140	48	48-13140	M	Beaumont-Port Arthur, TX CFS Area
16700	45	45-16700	M	Charleston-North Charleston, SC CFS Area
25540	09	09-25540	M	Hartford-East Hartford-Middletown, CT CFS Area
29700	48	48-29700	M	Laredo, TX CFS Area
40060	51	51-40060	M	Richmond, VA CFS Area
41740	06	06-41740	M	San Diego-Chula Vista-Carlsbad, CA CFS Area
45300	12	12-45300	M	Tampa-St. Petersburg-Clearwater, FL CFS Area
46520	15	15-46520	M	Urban Honolulu, HI CFS Area
47900	11	11-47900	M	Washington-Arlington-Alexandria, DC-VA-MD-WV CFS Area (DC Part)
	24	24-47900	M	Washington-Arlington-Alexandria, DC-VA-MD-WV CFS Area (MD Part)
	51	51-47900	M	Washington-Arlington-Alexandria, DC-VA-MD-WV CFS Area (VA Part)
99999	01	01-99999	R	Remainder of Alabama CFS Area
99999	02	02-99999	R	Remainder of Alaska CFS Area
99999	04	04-99999	R	Remainder of Arizona CFS Area
99999	05	05-99999	R	Remainder of Arkansas CFS Area
99999	06	06-99999	R	Remainder of California CFS Area
99999	08	08-99999	R	Remainder of Colorado CFS Area
99999	09	09-99999	R	Remainder of Connecticut CFS Area
99999	10	10-99999	R	Remainder of Delaware CFS Area
99999	12	12-99999	R	Remainder of Florida CFS Area
99999	13	13-99999	R	Remainder of Georgia CFS Area
99999	15	15-99999	R	Remainder of Hawaii CFS Area
99999	16	16-99999	R	Remainder of Idaho CFS Area
99999	17	17-99999	R	Remainder of Illinois CFS Area
99999	18	18-99999	R	Remainder of Indiana CFS Area
99999	19	19-99999	R	Remainder of Iowa CFS Area

Metro area of shipment	State of shipment	CFS Area of shipment	MA Type	Description
ORIG_MA DEST_MA	ORIG_STATE DEST_STATE	ORIG_CFS_AREA DEST_CFS_AREA		
99999	20	20-99999	R	Remainder of Kansas CFS Area
99999	21	21-99999	R	Remainder of Kentucky CFS Area
99999	22	22-99999	R	Remainder of Louisiana CFS Area
99999	23	23-99999	R	Remainder of Maine CFS Area
99999	24	24-99999	R	Remainder of Maryland CFS Area
99999	25	25-99999	R	Remainder of Massachusetts CFS Area
99999	26	26-99999	R	Remainder of Michigan CFS Area
99999	27	27-99999	R	Remainder of Minnesota CFS Area
99999	28	28-99999	R	Remainder of Mississippi CFS Area
99999	29	29-99999	R	Remainder of Missouri CFS Area
99999	30	30-99999	R	Remainder of Montana CFS Area
99999	31	31-99999	R	Remainder of Nebraska CFS Area
99999	32	32-99999	R	Remainder of Nevada CFS Area
99999	33	33-99999	R	Remainder of New Hampshire CFS Area
99999	35	35-99999	R	Remainder of New Mexico CFS Area
99999	36	36-99999	R	Remainder of New York CFS Area
99999	37	37-99999	R	Remainder of North Carolina CFS Area
99999	38	38-99999	R	Remainder of North Dakota CFS Area
99999	39	39-99999	R	Remainder of Ohio CFS Area
99999	40	40-99999	R	Remainder of Oklahoma CFS Area
99999	41	41-99999	R	Remainder of Oregon CFS Area
99999	42	42-99999	R	Remainder of Pennsylvania CFS Area
99999	45	45-99999	R	Remainder of South Carolina CFS Area
99999	46	46-99999	R	Remainder of South Dakota CFS Area
99999	47	47-99999	R	Remainder of Tennessee CFS Area
99999	48	48-99999	R	Remainder of Texas CFS Area
99999	49	49-99999	R	Remainder of Utah CFS Area
99999	50	50-99999	R	Remainder of Vermont CFS Area
99999	51	51-99999	R	Remainder of Virginia CFS Area
99999	53	53-99999	R	Remainder of Washington CFS Area
99999	54	54-99999	R	Remainder of West Virginia CFS Area
99999	55	55-99999	R	Remainder of Wisconsin CFS Area
99999	56	56-99999	R	Remainder of Wyoming CFS Area

The following codes only apply to shipment origin variables (and may appear as a result of Confidential Collapsing):

Metro area of shipment	State of shipment	CFS Area of shipment	MA Type	Description
ORIG_MA	ORIG_STATE	ORIG_CFS_AREA		
00000	NN	NN-00000		Origin metro area suppressed, origin state (NN) still shown
00000	00	00-00000		Origin completely suppressed

Notes:

(1) **MA Type** defined as:

C = Combined statistical area (CSA)-type CFS area

M = Metropolitan statistical area (MSA)-type CFS area

R = Remainder of state-type CFS area

Table A3.

Data Dictionary – SCTG Commodity Codes

SCTG	Description of commodity	SCTG Group
01	Animals and fish (live)	01-05
02	Cereal grains (includes seed)	
03	Agricultural products (excludes animal feed, cereal grains, and forage products)	
04	Animal feed, eggs, honey, and other products of animal origin	
05	Meat, poultry, fish, seafood, and their preparations	
06	Milled grain products and preparations, and bakery products	06-09
07	Other prepared foodstuffs and fats and oils	
08	Alcoholic beverages	
09	Tobacco products	
10	Monumental or building stone	10-14
11	Natural sands	
12	Gravel and crushed stone (excludes dolomite and slate)	
13	Other non-metallic minerals, not elsewhere classified	
14	Metallic ores and concentrates	
15	Coal	15-19
16	Crude petroleum	
17	Gasoline and aviation turbine fuel	
18	Fuel oils	
19	Other coal and petroleum products, not elsewhere classified	
20	Basic chemicals	20-24
21	Pharmaceutical products	
22	Fertilizers	
23	Other chemical products and preparations, not elsewhere classified	
24	Plastics and rubber	
25	Logs and other wood in the rough	25-30
26	Wood products	
27	Pulp, newsprint, paper, and paperboard	
28	Paper or paperboard articles	
29	Printed products	
30	Textiles, leather, and articles of textiles or leather	
31	Non-metallic mineral products	31-34
32	Base metal in primary or semi-finished forms and in finished basic shapes	
33	Articles of base metal	
34	Machinery	
35	Electronic and other electrical equipment and components, and office equipment	35-38
36	Motorized and other vehicles (includes parts)	
37	Transportation equipment, not elsewhere classified	
38	Precision instruments and apparatus	
39	Furniture, mattresses and mattress supports, lamps, lighting fittings, and illuminated signs	39-43
40	Miscellaneous manufactured products	
41	Waste and scrap	
43	Mixed freight	
00	Commodity code suppressed	

Notes:

- (1) SCTG 16 (Crude petroleum) is out of scope to the 2022 CFS.
- (2) Fields in the SCTG Group may appear as a result of Confidential Collapsing.